Gay-Lussac's law: how to isolate each quantity

I want p_2

1. Write the law. $$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$
2. The **T**s are **dividing**: each one goes to the other side **multiplying** (cross-multiply). $$p_1 \cdot T_2 = p_2 \cdot T_1$$
3. T_1 is **multiplying** p_2: divide **both sides** by T_1. $$\frac{p_1 \cdot T_2}{T_1} = \frac{p_2 \cdot \cancel{T_1}}{\cancel{T_1}}$$
4. ✓ Result: $$\boxed{p_2 = \frac{p_1 \cdot T_2}{T_1}}$$

I want T_2

1. Write the law. $$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$
2. The **T**s are **dividing**: each one goes to the other side **multiplying** (cross-multiply). $$p_1 \cdot T_2 = p_2 \cdot T_1$$
3. p_1 is **multiplying** T_2: divide **both sides** by p_1. $$\frac{\cancel{p_1} \cdot T_2}{\cancel{p_1}} = \frac{p_2 \cdot T_1}{p_1}$$
4. ✓ Result: $$\boxed{T_2 = \frac{p_2 \cdot T_1}{p_1}}$$

I want p_1

1. Write the law. $$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$
2. The **T**s are **dividing**: each one goes to the other side **multiplying** (cross-multiply). $$p_1 \cdot T_2 = p_2 \cdot T_1$$
3. T_2 is **multiplying** p_1: divide **both sides** by T_2. $$\frac{p_1 \cdot \cancel{T_2}}{\cancel{T_2}} = \frac{p_2 \cdot T_1}{T_2}$$
4. ✓ Result: $$\boxed{p_1 = \frac{p_2 \cdot T_1}{T_2}}$$

I want T_1

1. Write the law. $$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$
2. The **T**s are **dividing**: each one goes to the other side **multiplying** (cross-multiply). $$p_1 \cdot T_2 = p_2 \cdot T_1$$
3. p_2 is **multiplying** T_1: divide **both sides** by p_2. $$\frac{p_1 \cdot T_2}{p_2} = \frac{\cancel{p_2} \cdot T_1}{\cancel{p_2}}$$
4. ✓ Result: $$\boxed{T_1 = \frac{p_1 \cdot T_2}{p_2}}$$

Shortcut: what is **dividing** on one side goes to the other side **multiplying**, and what is **multiplying** goes to the other side **dividing**.